ADDITIONAL FILE 1

Stage-Resolved Transcriptomic Analysis Reveals Distinct Molecular Programs Associated with Liver Fibrosis Severity

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Contents: Figs. S1–S2; Tables S1–S6; Supplementary Methods.

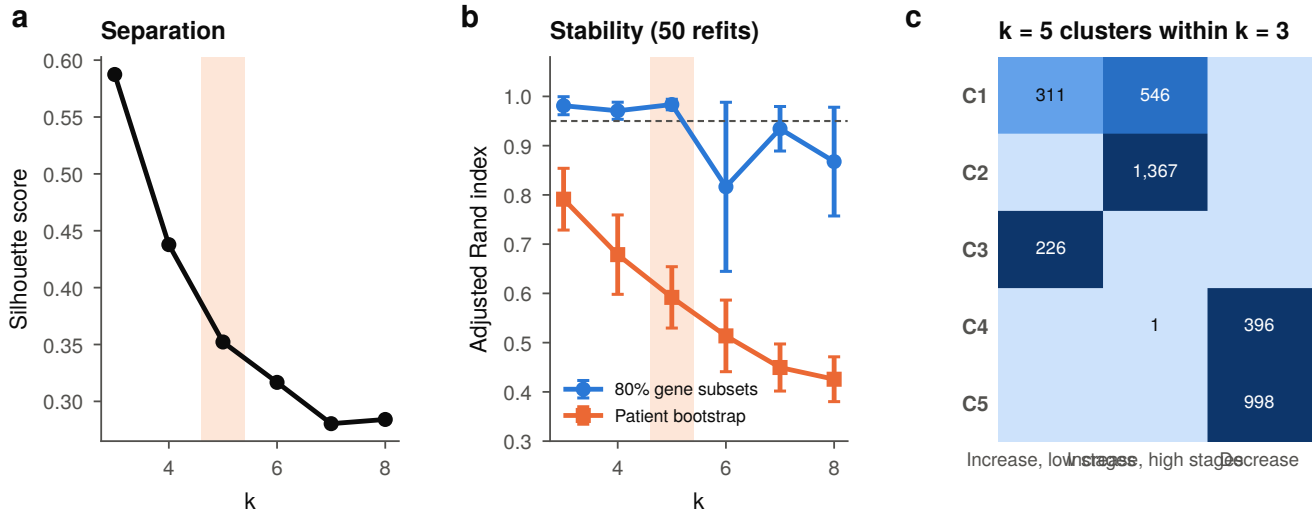


Figure S1. Choice of cluster number and nesting of the five patterns. (a) Silhouette score for $k = 3$–8. (b) Mean adjusted Rand index ($\pm$ SD) over 50 refits on 80% gene subsets and over 50 patient bootstraps; dashed line, 0.95 threshold; shaded, selected $k = 5$. (c) Cross-tabulation of the $k = 5$ clusters against the $k = 3$ solution recomputed with identical settings (50 initialisations, seed 42); $k = 3$ clusters are named from their centroid shapes. Re-running $k = 5$ reproduced the published labels exactly (ARI 1.000); 91.9% of genes fall in the $k = 3$ cluster containing most of their $k = 5$ cluster.

Table S1. Cluster-number diagnostics for all evaluated solutions. Eligibility was assessed on unrounded means.

k	Silhouette	Gene-subset ARI (mean $\pm$ SD)	Patient-bootstrap ARI (mean $\pm$ SD)	Meets rule
3	0.5874	0.9811 $\pm$ 0.0183	0.7913 $\pm$ 0.0627	Yes
4	0.4378	0.9708 $\pm$ 0.0174	0.6787 $\pm$ 0.0806	Yes
5	**0.3522**	**0.9835 $\pm$ 0.0107**	**0.5919 $\pm$ 0.0623**	**Yes (selected)**
6	0.3167	0.8164 $\pm$ 0.1717	0.5137 $\pm$ 0.0727	No
7	0.2804	0.9343 $\pm$ 0.0452	0.4495 $\pm$ 0.0478	No
8	0.2841	0.8676 $\pm$ 0.1104	0.4257 $\pm$ 0.0455	No

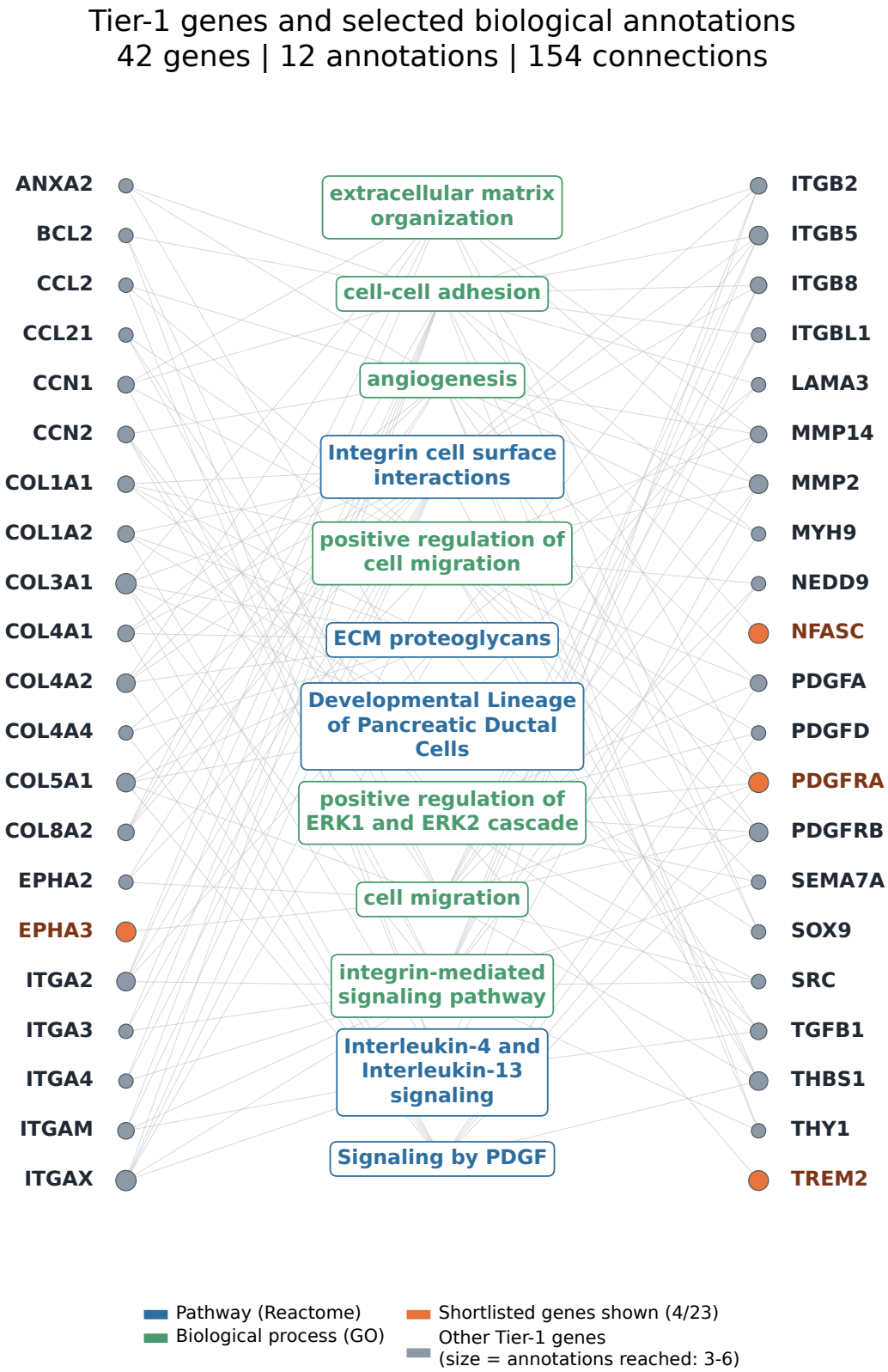


Figure S2. Knowledge-graph annotation subgraph of Tier 1 genes. Twelve non-redundant Pathway/BioProcess annotations over-represented among Tier 1 genes and the 42 Tier 1 genes linked to them through 154 curated relationships. Displayed genes reach at least three selected annotations or are shortlisted genes reaching at least one (*PDGFRA*, *EPHA3*, *NFASC*, *TREM2*). Links denote curated annotation membership, not physical or causal interactions.

Table S2. Dataset-specific preprocessing and final analysis eligibility. Counts are patient-level observations within each study; the two analysis populations overlap and must not be added. Zero indicates no contribution to that analysis, not exclusion from the expression resource.

Dataset	Expression processing	Sample handling and eligibility	Fibrosis n	Disease n
GSE126848	Raw counts; $\log_2$(CPM+1)	12 obese-only samples excluded from the ordered analysis; no fibrosis stages.	0	45
GSE130970	TPM; $\log_2$(TPM+1)	Five ambiguous classifications set to unclassified; remaining labels histology-derived.	78	73
GSE135251	Counts; $\log_2$(CPM+1)	Broad NAFLD labels excluded, leaving no variation on the ordered axis.	216	0
GSE162694	Raw counts; $\log_2$(CPM+1)	As GSE135251.	143	0
GSE167523	Raw counts; $\log_2$(CPM+1)	All 98 NAFL/NASH observations eligible; no fibrosis stages.	0	98
GSE193066	RLE-normalised counts; $\log_2$(count+1)	164 biopsies resolved to 106 patients (28 same-stage pairs averaged; 30 first biopsy).	106	0
GSE240729	Raw counts; $\log_2$(CPM+1)	All staged observations retained; no disease labels.	67	0
GSE269412	Raw counts; $\log_2$(CPM+1)	Expression resource only; required annotations unavailable.	0	0
Total			**610**	**216**

Table S3. Gene-prioritisation components and decision rules. $S_g = P_g + T_g + X_g + F_g + B_g$ (maximum 9). Tier 1, $S_g \geq 7$; Tier 2, $5 \leq S_g < 7$; Tier 3, $S_g < 5$. Values rounded to three decimals; lower thresholds inclusive. The components share evidence and are not independent validations.

Score	Evidence	Assignment rule
P_g	Association magnitude	2: $\lvert\text{median}\,\rho\rvert \geq 0.40$; 1: $0.30 \leq \lvert\text{median}\,\rho\rvert < 0.40$; 0 otherwise.
T_g	F0–F4 endpoint separation	$\Delta_g = \bar{z}_{F4} - \bar{z}_{F0}$ must share the sign of the median correlation; then 2: $\lvert\Delta_g\rvert \geq 1$; 1: $0.5 \leq \lvert\Delta_g\rvert < 1$; 0 otherwise.
X_g	Largest adjacent-stage contrast	For the contrast with the largest $\lvert\beta\rvert$ (ties in stage order): A, $q < 0.05$ and $\lvert\beta\rvert \geq 0.2$; B, $\geq 50\%$ of matched draws with $\lvert\beta\rvert \geq 0.2$ and the same sign. 2 if A and B; 1 if one; 0 if neither.
F_g	Fibrosis-only selection	1 if selected on the fibrosis axis only; 0 otherwise. Not a claim of biological specificity.
B_g	Functional annotation	Number of collections (GO BP, KEGG, Reactome, Hallmark) with a significant term containing the gene, via its direction group or full cluster. 2: ≥ 2; 1: one; 0: none.

Table S4. Composition of the integrated knowledge graph and drug-annotation coverage. Drug annotations were not inputs to the transcriptomic analyses.

Measure	Count
Total nodes / relationships	126,999 / 35,160,254
Broad entity categories / relationship types	8 / 15
KEGG (legacy) + Hallmark pathway nodes	236
Gene → KEGG/Hallmark pathway links (`IN_PATHWAY`)	20,031
Pathway → NAFLD links (`IMPLICATED_IN`)	89
ChEMBL indication links for added drugs (`TREATS`)	2,087
Expression relationships	32,471,042
Fibrosis / disease-group progression links	3,845 / 4,199
Drug nodes (OptimusKG-derived / added from ChEMBL, DGIdb, GtoPdb, Open Targets)	12,544 (12,025 / 519)
`ACTS_ON` relationships (OptimusKG-derived / four-database integration)	25,890 (20,674 / 5,216)
Common source-list / action-list fields populated	25,890 / 25,430
Four-database links involving the 519 added drugs	846
Four-database links involving OptimusKG drugs	4,370 (2,104 drugs)
Drug fields populated (of 12,544): SMILES / molecule type / max phase	9,036 / 12,051 / 10,293
Drug fields populated: approval / withdrawal flag / recovered name	9,902 / 11,623 / 10,878
OptimusKG drug–gene records with reference URLs / mechanism text (of 20,674)	2,893 / 2,898
Displayed annotation subgraph	42 genes; 12 annotations; 154 links

Drug sources: GtoPdb 2026.3.1; DGIdb 5 (October 2026); ChEMBL REST API (October 2026); Open Targets Platform 26.09. A populated approval or withdrawal flag may be true or false; a phase field does not establish approval or efficacy for liver disease.

Table S5. Deposited expression files.

Accession	File
GSE126848	GSE126848_Gene_counts_raw.txt.gz
GSE130970	GSE130970_all_sample_salmon_tximport_TPM_entrez_gene_ID.csv.gz
GSE135251	GSE135251_RAW.tar
GSE162694	GSE162694_raw_counts.csv.gz
GSE167523	GSE167523_Raw_gene_counts_matrix.txt.gz
GSE193066	GSE193066_NAFLD.HUn164.gct.gz
GSE240729	GSE240729_merged.counts.67samp.txt.gz
GSE269412	GSE269412_DRXOFLO_rnaseq_count_mat.txt.gz

Table S6. Biological context of the 23 shortlisted genes. Genes as in main-text Table 5, grouped by expression cluster. "Established biological function" summarises the general role of each gene product; "previous evidence in liver disease" indicates whether a liver-disease association is established; "stage pattern" describes the cluster's cross-sectional expression pattern across F0–F4 and does not imply a temporal sequence within patients. —, no specific prior evidence in liver fibrosis was identified for this table; these genes are candidates for follow-up rather than established fibrosis genes.

Gene	Cluster	Established biological function	Previous evidence in liver disease	Stage pattern in this study
IL32	C1	Inflammatory cytokine signalling	Upregulated in NAFLD liver; candidate circulating biomarker[a]	Higher from F1, increasing with stage
EPHA3	C1	Ephrin receptor signalling	—	Higher from F1, increasing with stage
LYZ	C1	Myeloid-associated immune function	—	Higher from F1, increasing with stage
NFASC	C1	Cell adhesion and organisation	—	Higher from F1, increasing with stage
PLPP4	C1	Lipid phosphate metabolism	—	Higher from F1, increasing with stage
PDGFRA	C2	PDGF receptor signalling	Fibrogenic mesenchymal biology	Increase, largest F3→F4
CACNA1C	C2	Calcium-channel signalling	—	Increase, largest F3→F4
TPM4	C2	Cytoskeletal organisation	—	Increase, largest F3→F4
PLK3	C2	Stress and cell-cycle regulation	—	Increase, largest F3→F4
TAX1BP3	C2	Signalling scaffold	—	Increase, largest F3→F4
TREM2	C3	Myeloid-cell-associated signalling	Scar-associated macrophage biology	Higher from F1, no further increase
CDC45	C3	DNA replication	Proliferation-associated biology	Higher from F1, no further increase
HTR7	C3	Serotonin receptor signalling	—	Higher from F1, no further increase
FABP5	C3	Fatty-acid binding and lipid handling	—	Higher from F1, no further increase
TYMS	C3	Nucleotide biosynthesis	Proliferation-associated biology	Higher from F1, no further increase
OAT	C4	Amino-acid metabolism	—	Decrease, largest F1→F2
KNG1	C4	Coagulation and inflammatory pathways	—	Decrease, largest F1→F2
F2	C4	Coagulation	Hepatic synthesis-associated biology	Decrease, largest F1→F2
SULT1E1	C5	Steroid metabolism	Hepatic metabolic function	Decrease, largest F3→F4
SLC38A4	C5	Amino-acid transport	Hepatic transport function	Decrease, largest F3→F4
CYP2C19	C5	Xenobiotic metabolism	Hepatic drug metabolism	Decrease, largest F3→F4
ACADSB	C5	Branched-chain amino-acid and fatty-acid metabolism	—	Decrease, largest F3→F4
MME	C5	Membrane metalloprotease	—	Decrease, largest F3→F4

[a] Baselli GA, et al. Liver transcriptomics highlights interleukin-32 as novel NAFLD-related cytokine and candidate biomarker. Gut. 2020;69(10):1855–66.

Supplementary Methods

Implementation details supplementing the main-text Methods.

Implementation details

Processing order. Duplicate raw identifiers were resolved (first occurrence retained); identifiers were mapped to Ensembl and unmappable or PAR-Y records excluded; distinct identifiers mapping to the same sample–gene pair were summed; the study-appropriate transformation was applied (CPM library totals computed before mapping and filtering); each gene was standardised within study; and patient-level biopsy selection or averaging followed. Equal-stage repeat biopsies were averaged on the standardised scale without restandardisation.

Implementation checks for scoring. For X_g, a missing full-data q-value did not satisfy the substantial-difference condition, and missing matched evidence halted scoring rather than defaulting to zero. In the final analysis all 3,845 fibrosis-associated genes had finite coefficients and q-values for all four contrasts and matched evidence for the selected contrast. Independently reconstructed P/T/X/F/B scores agreed with the analysis table for all 3,845 genes.

Graph verification. All 8,044 progression relationships and their association properties were compared with the analysis exports, and the 154 displayed gene–annotation connections were checked by endpoint identifiers and relationship type.

Reference-file provenance. Gene mapping used `hgnc_complete_set.txt` (file dated 20 August 2026; most recent record modification 18 August 2026); the knowledge layer used the OptimusKG parquet files (dated 1 September 2026; source deposit doi:10.7910/DVN/IYNGEV); functional annotation used MSigDB 2024.1.Hs. Formal release identifiers and expression-data download dates were not recorded; SHA-256 checksums of the retained files are provided in `source_checksums.txt`.

Reference-file SHA-256 checksums

The accompanying `source_checksums.txt` contains the reference-file manifest recorded on 21 October 2026 after the analysis. The entries below reproduce that record for the HGNC dictionary and 14 OptimusKG parquet files. Paths are relative to the analysis input directory; filesystem modification dates and hashes identify retained files, not official release dates. This manifest does not cover the subsequently added MSigDB, CTD or ChEMBL inputs.

```
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